



Institutional Review Board

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CONSENT TO PARTICIPATE

Project Title	<i>Multimodal Adversarial Question Answering</i>
Purpose of the Study	<i>This research is being conducted by Jordan Boyd-Graber at the University of Maryland, College Park. We are inviting you to participate in this research project because you are interested in answering audio quizbowl questions and 18 years of age or older. The purpose of this research project is to improve the models with audio-language reasoning capabilities.</i>
Procedures	Upon creating an account on the Earudite platform, you will have the ability to answer audio-based quizbowl questions across various categories. The study will run from September 2024 to November 2024, during which you can access the platform at any time. We estimate that you will spend between 2 and 30 hours participating over the course of the study, with daily sessions likely ranging from 10 to 60 minutes. There is no minimum time commitment required. You will choose a username which will be made public on leaderboards, so please select it carefully to maintain anonymity. To begin, you will be presented with audio questions to answer. You will have the option to submit your answers either by speaking (which will be recorded as an audio response) or by typing text. If you choose to give audio responses, you will need to ensure you have a working microphone available. The platform will record information including your answer, its correctness, the point at which you buzzed in, and your user ID. This data will be stored securely in our database. The only identifying information tied to your user ID is your email and username. A leaderboard will display all users who have completed some number of questions, showing their username and "knowledge score." The knowledge score reflects your ability to answer the audio trivia questions correctly and quickly. Once you've answered a question, it won't be presented to you again. Your participation in this research is a requirement for using the Earudite platform. However, you can choose to stop participating at any time without penalty.
Potential Risks and Discomforts	The non-physical known risk to you is in the confidentiality of your study participation and data from answering questions with the app. This risk will be mitigated through the procedures described in the Confidentiality section. Physical risks could include potentially loud sounds within the audio files, which can be unexpected and

	troublesome for people who have sensitive hearing. Also, there may be themes mentioned within the audio samples that could be potentially triggering, which include violence, sexual acts, or the use of swear words.
Potential Benefits	The benefits to you include the opportunity to engage with a wide range of audio-based trivia questions on the Earudite platform, enhancing your knowledge across various categories. While there are no other direct benefits, this study aims to create a human benchmark for audio-based trivia questions, which could lead to improvements in both human and AI performance in this area. By participating, you contribute to the advancement of research in audio processing, natural language understanding, and question-answering systems.
Confidentiality	Data collected in this study will be securely stored in a database hosted on password-protected web servers and cloud-based file storage. Only researchers will have access to your email, and no other identifying information will be collected through the platform. Regarding the public leaderboard, only your chosen username will be displayed. Therefore, you will be able to remain anonymous on the leaderboard if you desire. Since emails are classified as PII (personally identifiable information), they will not be included on the public leaderboard. After the end of the experiment, data will be anonymized and released publicly. User IDs are used throughout the experiment to serve questions to you and produce statistical leaderboards. The collected usernames, emails, and links to public statistics pages will be destroyed after the completion of this study. To maintain linkage in study data, a new randomized user ID will be generated in replacement of the identification keys used during the study. We also use cookies to store certain user data, which is also present in our database. The cookies are stored on your browser and can be deleted by you at any time.
Compensation	At the end of the study, compensation will be distributed as follows: The top five (5) participants with the highest knowledge scores and the top five (5) participants with the highest number of questions attempted will each receive a \$30 bonus. Additionally, we will implement a tiered lottery system. For every 50 questions completed (up to 250 questions), participants will be entered into a separate lottery draw. Each tier (50, 100, 150, 200, 250 questions) will have its own draw where ten (10) people will win \$10 gift cards. Participants can be eligible for multiple tiers based on their total questions answered, potentially winning in multiple draws. To ensure broader participation benefits, the first 100 participants who complete 25 or more questions will be guaranteed to receive \$5. All monetary rewards will be distributed via gift cards sent to the email address associated with your account. While we expect between 50-1,000 entrants, we will allow up to 100,000 entrants. Providing

	blatantly nonsensical or irrelevant answers may deem you ineligible for prizing. Please note that the number of guaranteed compensations and lottery winners is fixed, regardless of the total number of participants. All the compensations will be paid out in a secure and virtual way via Tango platform.
Right to Withdraw and Questions	Your participation in this research is completely voluntary. You may choose not to take part at all. If you decide to participate in this research, you may stop participating at any time. If you decide not to participate in this study or if you stop participating at any time, you will not be penalized or lose any benefits to which you otherwise qualify. If you decide to stop taking part in the study, if you have questions, concerns, or complaints, or if you need to report an injury related to the research, please contact the investigator: Jordan Boyd-Graber 4146 Iribe University of Maryland College Park, Maryland, 20742 E-mail: jbg@umiacs.umd.edu Telephone: (301) 405-6766
Participant Rights	<i>If you have questions about your rights as a research participant or wish to report a research-related injury, please contact:</i> University of Maryland College Park Institutional Review Board Office 1204 Marie Mount Hall College Park, Maryland, 20742 E-mail: irb@umd.edu Telephone: 301-405-0678 <i>For more information regarding participant rights, please visit:</i> https://research.umd.edu/research-resources/research-compliance/institutional-review-board-irb/research-participants <i>This research has been reviewed according to the University of Maryland, College Park IRB procedures for research involving human subjects.</i>
Statement of Consent	<i>By proceeding and clicking the button below you are indicating that you are at least 18 years of age; you have read this consent form or have had it read to you; your questions have been answered to your satisfaction and you voluntarily agree to participate in this research study.</i>